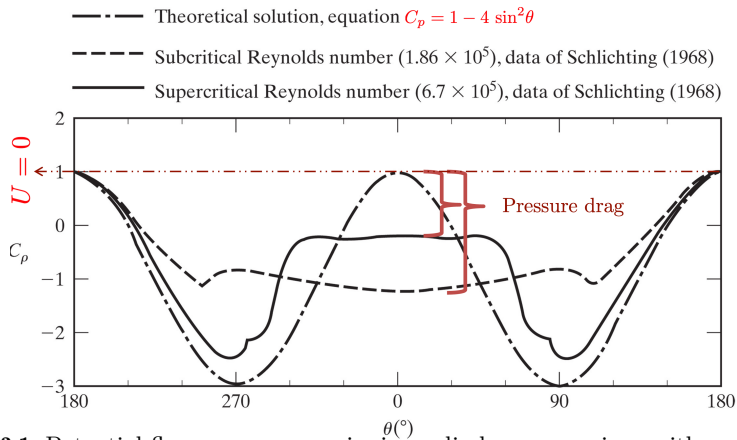


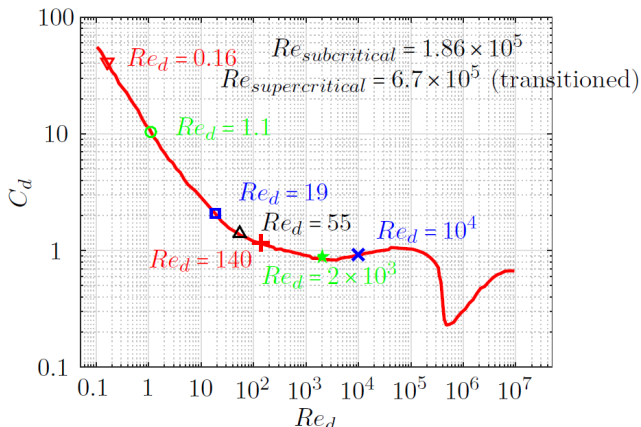
# Introduction to Boundary Layers (Chapter 3 & 4)

# Flow over a non-spinning cylinder



**Figure 3.1:** Potential flow over a non-spinning cylinder, comparison with experiments, modified Fig. 3.17 from AE-BC

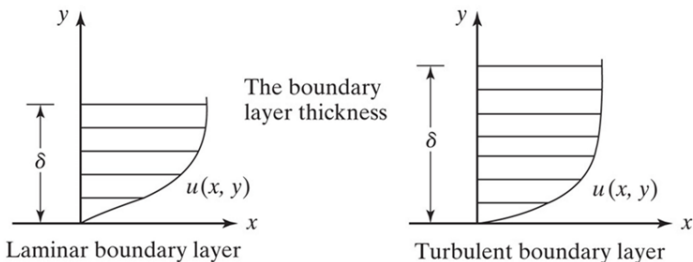
# Different flow regimes



**Figure 3.2:** Variation of Drag Coefficient  $C_d$  of a cylinder as a function of the Reynolds number  $Re_d$

Need to understand, boundary layers, pressure gradients and flow separation to explain.

# Laminar & Turbulent BL

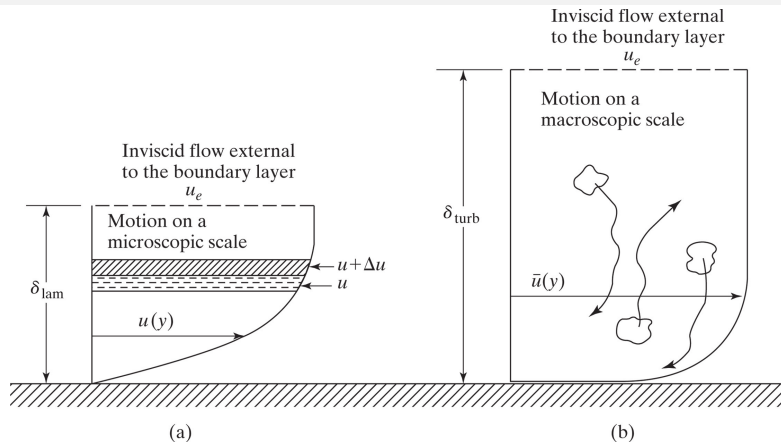


**Figure 3.3:** Laminar and turbulent boundary layer, Fig 2.14 from AE-BC

- ▶ Boundary layers arise from the no-slip boundary condition
- ▶ Two broad types - laminar and turbulent
- ▶ In general, turbulent boundary thickness  $\delta_{turb} >$  laminar boundary thickness  $\delta_{lam}$
- ▶ Note  $\delta$  also referred to as  $\delta_{99}$  is the  $z$  location such that  $u(z = \delta) = 0.99U_e$



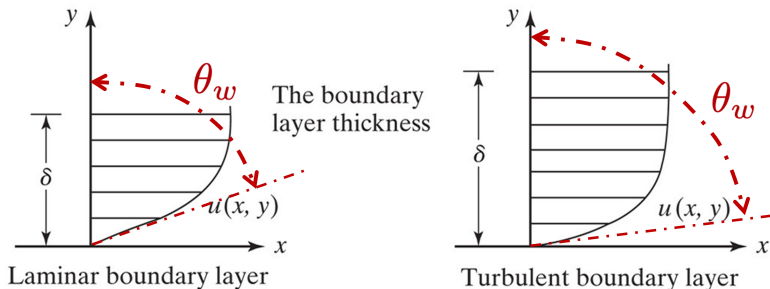
# Laminar & Turbulent BL



**Figure 3.4:** Laminar and turbulent boundary layer, Fig 4.2 from AE-BC

- ▶ Laminar → move in lamina or layers
- ▶ Turbulent → large-scale mixing - brings more momentum closer to the wall resulting in a fuller velocity profile

# Laminar & Turbulent BL

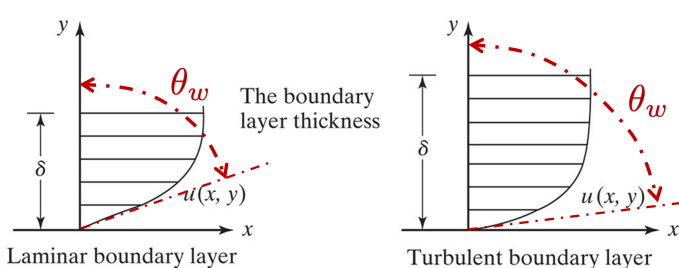


**Figure 3.5:** Wall shear stress, modified Fig 2.14 from AE-BC

The wall shear stress or skin friction is given by:

$$\tau_w = \mu \left[ \frac{\partial u}{\partial y} \right]_{y=0} \propto \left[ \frac{\Delta u}{\Delta y} \right]_{y=0} \approx \tan \theta_w \quad (3.1)$$

# Laminar & Turbulent BL



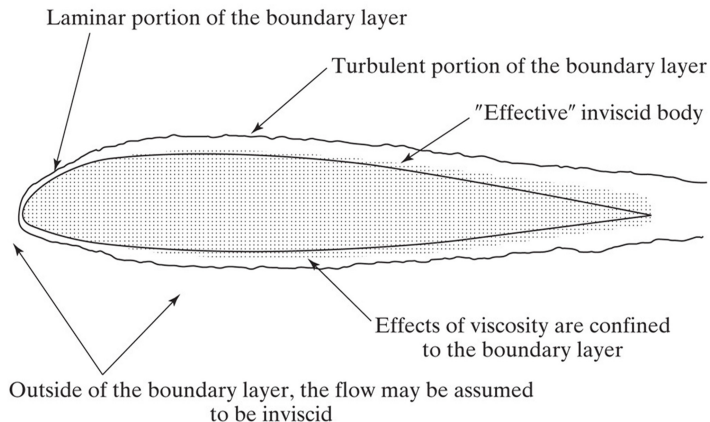
**Figure 3.6:** Wall shear stress, modified Fig 2.14 from AE-BC

**Turbulent** boundary layers have greater  $[\partial u / \partial y]_{y=0}$  than **laminar** boundary layers. Hence, **turbulent** boundary layers have great wall shear stress and thus skin friction drag. Note,

$$C_{D\ sf} \propto C_f \frac{S_{wetted}}{S} \quad (3.2)$$

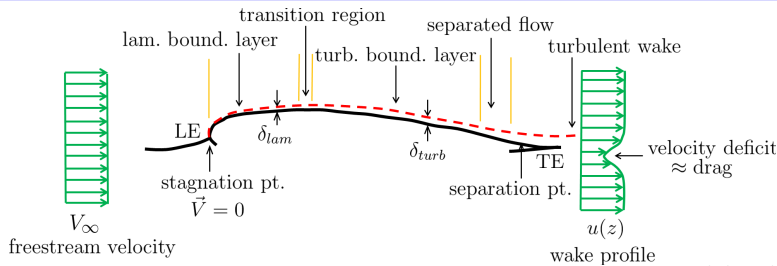
where  $S_{wetted}$  is the area “wetted” by the fluid and  $C_{D\ sf}$  the coefficient of drag due to skin friction

# Boundary layer over an airfoil



**Figure 3.7:** Boundary layer over an airfoil, Fig 2.14 from AE-BC

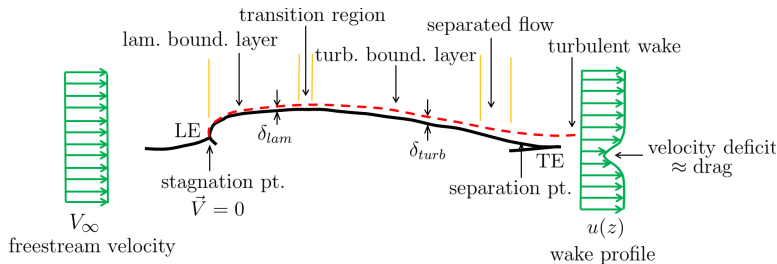
# Boundary layer over an airfoil



**Figure 3.8:** Boundary layers over airfoils partially adapted from AA-DA

- ▶ The boundary layer generally starts as **laminar** close to the LE at the stagnation point where  $\vec{V} = 0$
- ▶ The stagnation point will vary with  $\alpha$
- ▶ Only the top surface is shown in Fig. 3.10, similar behavior is seen on the bottom surface

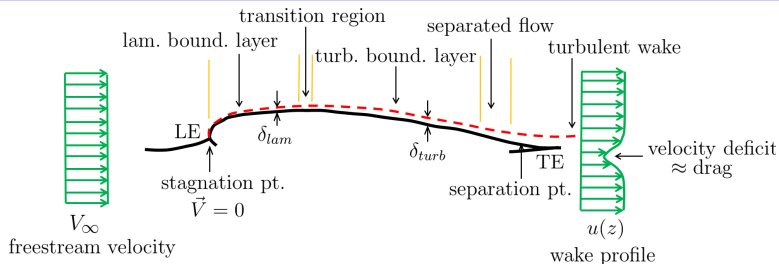
# Boundary layer over an airfoil



**Figure 3.9:** Boundary layers over airfoils partially adapted from AA-DA

- ▶ The boundary layer then grows in thickness and at some point transitions to a turbulent boundary layer over a region known as the **transition region**
- ▶ The turbulent boundary layer then grows in thickness also and might separate

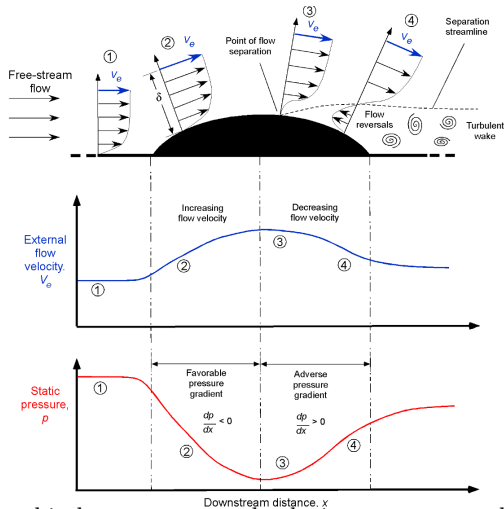
# Boundary layer over an airfoil



**Figure 3.10:** Boundary layers over airfoils partially adapted from AA-DA

- ▶ Due to no slip the boundary layer **removes** momentum from the freestream velocity which shows up as the velocity deficit in the wake. This velocity deficit is proportional to the drag.

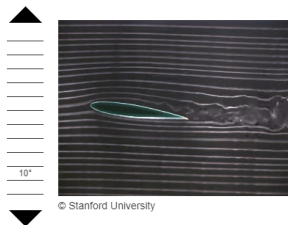
# Flow separation effects



**Figure 3.11:** Relationship between external velocity, pressure gradient and flow separation. From IAFV-JGL



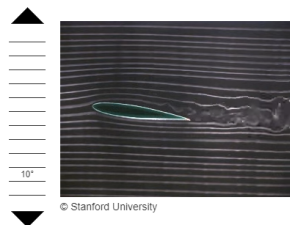
# Flow separation over an airfoil



**Figure 3.12:** Flow separation over an airfoil, From MFM

- ▶ Over the leading edge of an airfoil the flow is accelerating which means  $U_e$  is **increasing** and the boundary layer profile becomes “fuller” while  $p_e$  is **decreasing** – **favorable pressure gradient**  $\partial p_e / \partial s < 0$ .
- ▶ Over the aft of the airfoil the flow is decelerating which means  $U_e$  is **decreasing** and the boundary layer profile becomes less and less “full” while  $p_e$  is **increasing** – **adverse pressure gradient**  $\partial p_e / \partial s > 0$ .

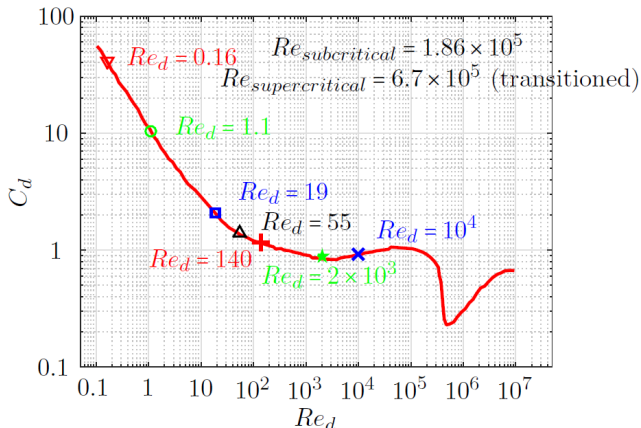
# Flow separation over an airfoil



**Figure 3.13:** Flow separation over an airfoil

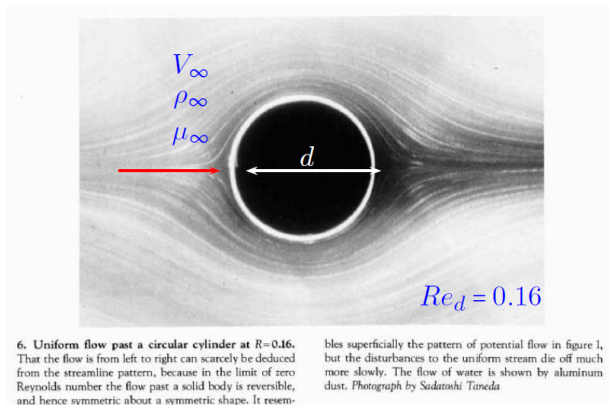
- ▶ If the adverse pressure gradient is sufficiently strong then the flow will **separate (reversed flow)** as shown
- ▶ It is clear (refer to Fig. 3.3) that a **turbulent** boundary layer is less likely to separate than a **laminar** boundary layer as the **turbulent** boundary layer has **more momentum closer to the wall**
- ▶ Within a lam.b.l.  $\partial p / \partial z = 0$  and for turb.b.l  $\partial p / \partial z \approx 0$ . Therefore,  $p(z = \delta) = p_e = p(z = 0) \rightarrow p$  is constant along a normal to the wall

# Different flow regimes



**Figure 3.14:** Variation of Drag Coefficient  $C_d$  of a cylinder as a function of the Reynolds number  $Re_d$

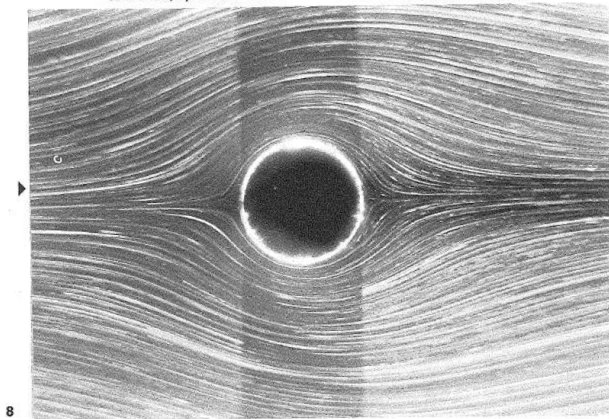
# $C_d$ of a cylinder



**Figure 3.15:** Visualization of flow over a cylinder at  $Re_d = 0.16$  from AFM-VD. (▽ in figure 3.14)

## $C_d$ of a cylinder

Fig. 2. Flow around a circular cylinder at  $Re = 1.1$  (glycerine–water solution, flow velocity 0.20 cm/s, cylinder diameter 1.0 cm, aluminium powder method).



**Figure 3.16:** Visualization of flow over a cylinder at  $Re_d = 1.1$  from VF-JSME. (○ in figure 3.14)

## $C_d$ of a cylinder

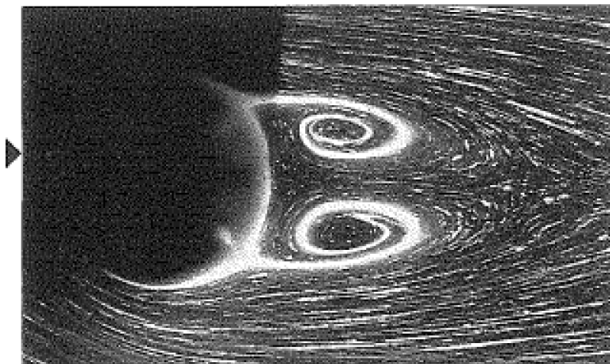


Fig. 3. Flow around a circular cylinder at  $Re = 19$  (water, flow velocity 0.20 cm/s, cylinder diameter 1.0 cm, aluminium powder method and electrolytic precipitation method).

**Figure 3.17:** Visualization of flow over a cylinder at  $Re_d = 19$  from VF-JSME.

## $C_d$ of a cylinder

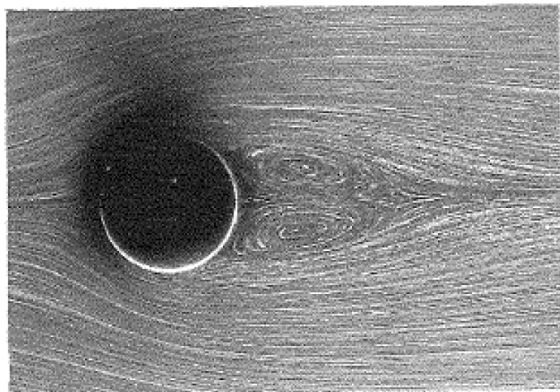


Fig. 4. Flow around a circular cylinder at  $Re = 26$  (water, flow velocity 0.25 cm/s, cylinder diameter 1.0 cm, aluminium powder method).

**Figure 3.18:** Visualization of flow over a cylinder at  $Re_d = 26$  from VF-JSME.

## $C_d$ of a cylinder

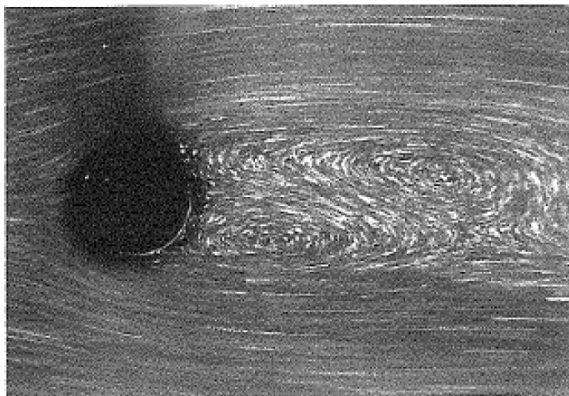


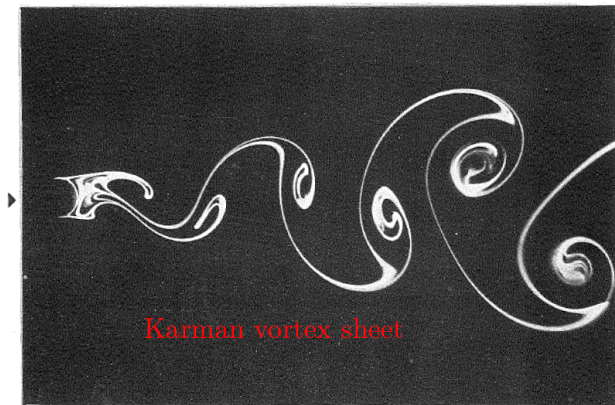
Fig. 5. Flow around a circular cylinder at  $Re = 55$  (water, flow velocity 0.55 cm/s, cylinder diameter 1.0 cm, aluminium powder method).

**Figure 3.19:** Visualization of flow over a cylinder at  $Re_d = 55$  from VF-JSME. ( $\Delta$  in figure 3.14)



## $C_d$ of a cylinder

Fig. 6. Karman vortex street behind a circular cylinder at  $Re = 140$  (water, flow velocity 1.5 cm/s, cylinder diameter 1.0 cm, electrolytic precipitation method).



**Figure 3.20:** Visualization of flow over a cylinder at  $Re_d = 140$  from VF-JSME. (+ in figure 3.14)

## $C_d$ of a cylinder

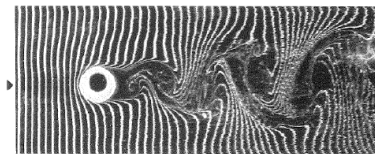


Fig. 7. Time lines.

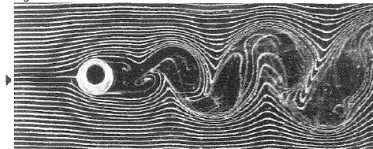


Fig. 8. Streak lines.

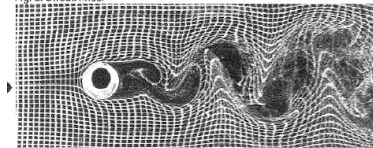
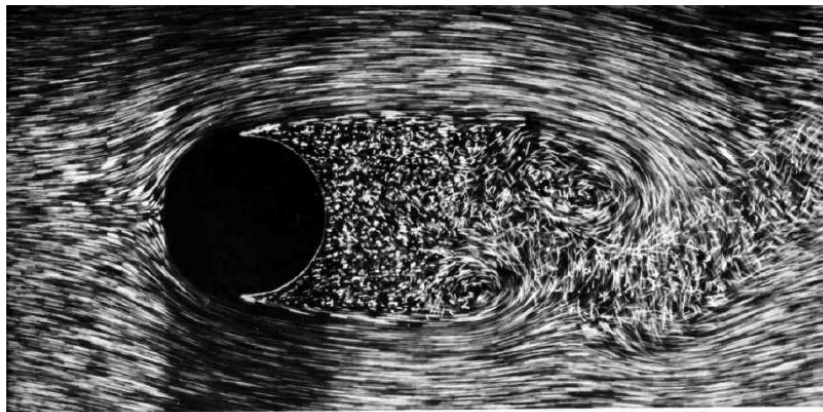


Fig. 9. Time lines and streak lines.

Figs. 7–9. Kármán vortex street behind a circular cylinder at  $Re = 170$  (water, flow velocity 2.6 cm/s, cylinder diameter 8 mm, hydrogen bubble method).

**Figure 3.21:** Visualization of flow over a cylinder at  $Re_d = 170$  from VF-JSME.

## $C_d$ of a cylinder

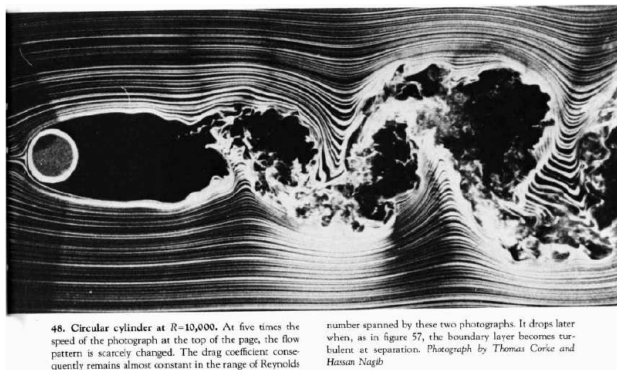


47. Circular cylinder at  $R=2000$ . At this Reynolds number one may properly speak of a boundary layer. It is laminar over the front, separates, and breaks up into a turbulent wake. The separation points, moving forward as

the Reynolds number is increased, have now attained their upstream limit, ahead of maximum thickness. Visualization is by air bubbles in water. ONERA photograph, Werlé & Gallon 1972

**Figure 3.22:** Visualization of flow over a cylinder at  $Re_d = 2000$  from AFM-VD. (★ in figure 3.14)

## $C_d$ of a cylinder



**Figure 3.23:** Visualization of flow over a cylinder at  $Re_d = 10000$  from AFM-VD. (× in figure 3.14)

## $C_d$ of a cylinder

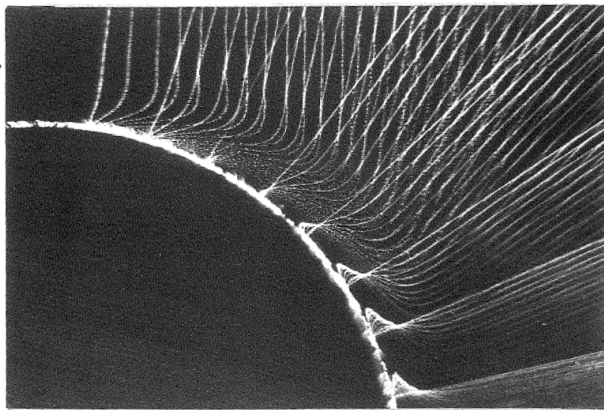
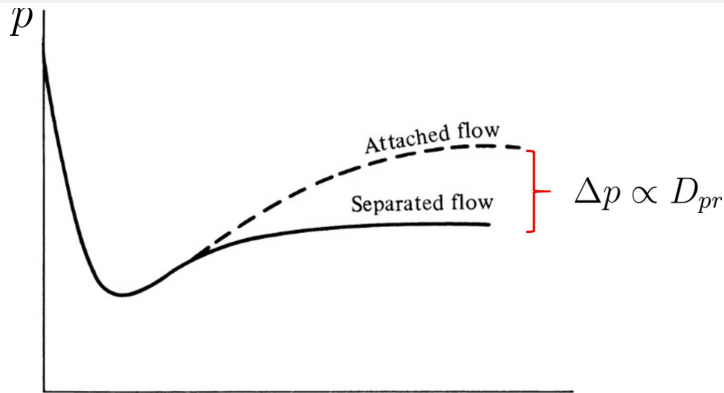


Fig. 22. Water, velocity of motion 2 cm/s, cylinder diameter 70 mm, photographed two seconds after the start of motion,  $Re = 1.2 \times 10^3$ , hydrogen bubble method.

**Figure 3.24:** Visualization of flow over a cylinder showing separation of the boundary layer at  $Re_d = 1.2 \times 10^3$  from VF-JSME.

# Pressure drag



**Figure 3.25:** Pressure drag origin adapted from Fig 15.4 from FA-JA

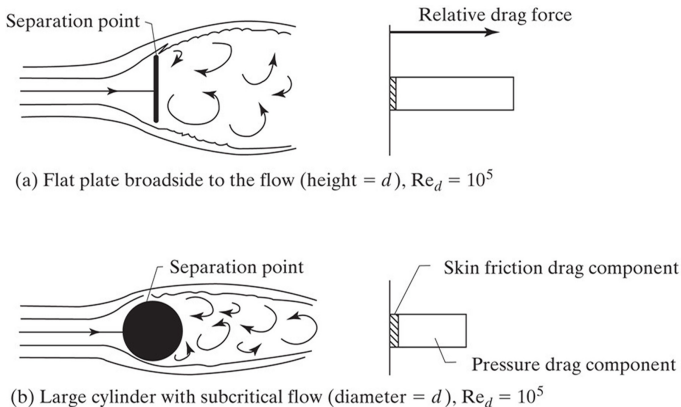
- ▶ When the flow is separated, a large amount of energy is being extracted from the free stream to “turn” the flow, this shows up as a pressure loss which is directly proportional to the pressure drag  $D_{pr}$

# Types of drag

So far we have learned about two types of drag, with a few more to come as we go along.

- ▶ pressure drag or form drag (as it depends on the shape of the body) -  $D_{pr}$  and the corresponding coefficient  $C_{D\ pr}$
- ▶ skin friction drag -  $D_{sf}$  and the corresponding coefficient  $C_{D\ sf}$
- ▶ If a body is a streamline body -  $D_{pr} \approx D_{sf}$
- ▶ If a body is a bluff body -  $D_{pr} \gg D_{sf}$

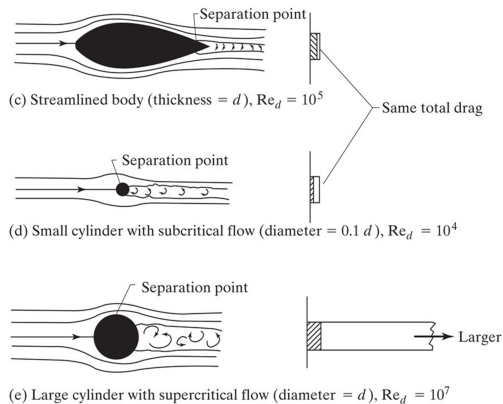
# Streamline and Bluff



**Figure 3.26:** Comparison of drag coefficients of various objects, Fig. 3.23 from AE-BC



# Streamline and Bluff

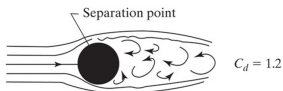


**Figure 3.27:** Comparison of drag coefficients of various objects, Fig. 3.23 from AE-BC

# Streamline and Bluff



(a) Flat plate broadside to the flow (height =  $d$ ),  $Re_d = 10^5$



(b) Large cylinder with subcritical flow (diameter =  $d$ ),  $Re_d = 10^5$



(c) Streamlined body (thickness =  $d$ ),  $Re_d = 10^5$

**Figure 3.28:** Comparison of drag coefficients of various objects, Fig. 3.23 from AE-BC